Generative Adversarial Minority Oversampling for Spectral–Spatial Hyperspectral Image Classification

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***Abstract*— Recently, convolutional neural networks (CNNs) have exhibited commendable performance for hyperspectral image (HSI) classification. Generally, an important number of samples are needed for each class to properly train CNNs. However, existing HSI data sets suffer from a significant class imbalance problem, where many classes do not have enough samples to characterize the spectral information. The performance of existing CNN models is biased toward the majority classes, which possess more samples for the training. This article addresses this issue of imbalanced data in HSI classification. In particular, a new `3D-HyperGAMO` model is proposed, which uses generative adversarial minority oversampling. The proposed `3D-HyperGAMO` automatically generates more samples for minority classes at training time, using the existing samples of that class. The samples are generated in the form of a 3-D hyperspectral patch. A different classifier from the generator and the discriminator is used in the `3D-HyperGAMO` model, which is trained using both original and generated samples to determine the classes of newly generated samples to which they actually belong. The generated data are combined classwise with the original training data set to learn the network parameters of the class. Finally, the trained 3-D classifier network validates the performance of the model using the test set. Four benchmark HSI data sets, namely, Indian Pines (IP), Kennedy Space Center (KSC), University of Pavia (UP), and Botswana (BW), have been considered in our experiments. The proposed model shows outstanding data generation ability during the training, which significantly improves the classification performance over the considered data sets. The source code is available publicly at https://github.com/mhaut/3D-HyperGAMO.**

***Index Terms*— Convolutional neural networks (CNNs), deep learning, spectral–spatial hyperspectral image (HSI) classification.**

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I. Introduction

HYPERSPECTRAL images (HSIs) capture the spectral information reflected by materials by gathering different wavelengths along the electromagnetic spectrum in hundreds of narrow, contiguous bands. The resulting data cubes provide valuable information about the observed materials, where each pixel represents the spectral signature of an imaged object. Resulting from the rich information contained in the data, HSI analysis has been successfully conducted in several applications related to Earth observation and remote sensing, such as vegetation modeling, urbanization analysis, crop analysis, and many more [1]. Several attempts have been conducted for HSI classification using classical pattern recognition and machine learning methods, such as kernel-based methods [2], composite kernels [3], dictionary-based sparse representation [4], multiple feature learning [5], fusion of correlation coefficient, and joint sparse representation [6]. However, these approaches usually depend on handcrafted features, which are difficult to obtain and hinged on expert knowledge.

A. Brief Review of the Current State of the Art Within Deep Learning and HSI Classification: Milestones and Limitations

In recent years, traditional feature representation and learning methods have been superseded by convolutional neural networks (CNNs) [7]. The CNN is a hierarchical feature learning and classification technique whose architecture is composed of a stack of different layers, including convolutions and activation functions [8]. In this sense, the bottom layers provide simple and low-abstract features from the input data, which are further sophisticated as they are processed by deeper layers, until the top layers obtain very representative and abstract features. In this way, the CNN extracts the most relevant features automatically from the training data, exhibiting outstanding results in a wide range of computer vision applications [9] and text analysis [10], among others. In particular, CNNs have shown great performance in remote sensing HSI classification problems. Many CNN-based models are currently available in the HSI classification literature. For instance, some papers combine CNN models with handcrafted features, such as He *et al.* [11] who consider covariance

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matrices and CNN models. However, the vast majority of methods consider features automatically extracted by CNN models. In this context, an optimized CNN is proposed by Yu *et al.* [12], which applies three 1×1 convolutional layers with average pooling layer and larger dropout rates over $5 \times 5 \times B$ input HSI patches (here, B denotes the number of bands), reducing the spectral dimension while keeping constant the spatial dimension as the HSI feature volumes go deep through the model. Also, Paoletti *et al.* [13] developed CNNs with three convolutional layers considering different input spatial patches and kernel sizes, by expanding the spectral domain and then reducing it to extract the most relevant spectral information while combining it with the contextual one. Makantasis *et al.* [14] have implemented a CNN2D for HSI classification, introducing randomized principal component analysis (R-PCA) [15] to extract the first 10–30 principal components in order to reduce the computational cost of the training and prediction stages. Furthermore, Li *et al.* [16] have exploited the learning of the pixel-pair features of deep CNN models to increase their discriminative ability by combining the HSI patch center pixel with each of its neighboring pixels, applying a voting strategy at the end.

Also, more complex strategies have been combined with CNN-based models in order to enhance their feature extraction (FE) ability. For instance, Haut *et al.* [17] and Cao *et al.* [18] applied active learning to obtain the most discriminative features to enhance the classification performance. Currently, residual learning [19] has gained a lot of popularity since a wide variety of models and architectures have been successfully applied within HSI data classification problems. For instance, Kang *et al.* [20] have developed a dual-path CNN for HSI classification by combining different levels of features with residual learning. Also, Song *et al.* [21] have employed deep feature fusion with residual connections for the same purpose. Moreover, Cheng *et al.* [22] have explored different kinds of features for HSI classification. Hamida *et al.* [23] have employed the CNN3D for HSI classification, which has been further extended by He *et al.* [24] to multiscale CNN3D. Furthermore, Zhong *et al.* [25] have used the 3-D residual network, developing the spectral–spatial residual network (SSRN). Paoletti *et al.* [26] have proposed a new residual pyramid bottleneck to keep the computational loads balanced along the model, elongating the spectral domain while reducing the spatial shape of obtained features. Also, Paoletti *et al.* [27] extended the application of previous features by developing a densely connected residual model. Moreover, several works, such as by Haut *et al.* [28], Fang *et al.* [29], Xu *et al.* [30], Wu *et al.* [31], Alipour-Fard [32], and Gao *et al.* [33], combined residual connections with attention mechanism to enhance the FE process, highlighting the most useful features for HSI data classification and attenuating the less relevant ones. Recently, Roy *et al.* [34] have proposed the HybridSN to utilize both spectral and spatial features by sequentially combining 3-D and 2-D convolutions and lightweight spectral–spatial squeeze-and-excitation residual bag-of-features learning to reduce the trainable parameters for HSI classification [35]. Moreover, Zheng *et al.* [36] have developed a fast patch-free global learning CNN for HSI classification. Liu *et al.* [37] have introduced content-guided CNNs for HSI classification.

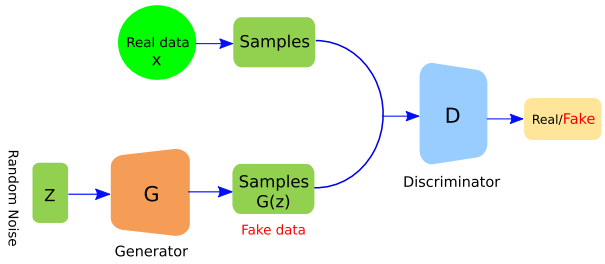


Fig. 1. GAN framework using generator and discriminator networks.

However, despite the outstanding results obtained by these and other popular deep models for HSI data interpretation, none of these works are able to tackle the fundamental problem of class imbalance, which is often present in HSI data sets. Indeed, this problem is relevant in classification tasks, as sample labeling is quite difficult and expensive in practice, while it is absolutely necessary to model the minority classes in order to conduct a complete and accurate land cover analysis. In this context, deep models are usually biased to learn the information from large-sized classes while disregarding the small-sized ones [38], [39]. As a result, majority classes have a higher impact on the classification performance, which provides an unrealistic (enhanced) overall accuracy (OA), while the model performs poorly on minority classes, hindering the real performance given by metrics such as the average accuracy (AA) and the class-specific accuracy [40], [41].

Several approaches have been suggested to deal with the aforementioned class imbalance problem [42], [43]. For instance, sampling-based methods are quite popular in this context due to their simplicity. They preprocess the data set through artificial resampling, which is done by undersampling the majority classes (i.e., removing some samples from the majority class samples, for instance, via random selection) [44], [45], by oversampling the minority classes (i.e., increasing the number of minority class samples, for instance, with data augmenting or sample replication) [46], [47], or by a combination of both approaches [48].

B. Tackling the Class Imbalance Problem With GANs

In addition to convolution-based models and residual connections, the deep learning field has been revolutionized by the introduction of a new type of architecture, known as generative adversarial networks (GANs). GANs were originally proposed by Goodfellow *et al.* [49] in order to facilitate the generation of new samples in the given probability distribution. GANs use two networks, namely, generator and discriminator, as depicted in Fig. 1. The generator network provides new samples (i.e., fake samples) from a random noise vector, while the discriminator network recognizes the real and fake samples. The whole GAN is trained in such a way that the generator tries to fool the discriminator by generating more realistic samples, and the discriminator tries not to get fooled by the generator by classifying the generated samples as fake ones. Thus, a kind of adversarial learning (such as a minimax game) is performed to train both the generator and discriminator

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networks. Very recently, generative adversarial learning has been applied to remotely sensed HSI data to perform several tasks, such as image synthesis [50], data semantic segmentation [51], super-resolution [52], and anomaly detection [53], among others.

In the last three years, a significant number of works have been reported for HSI classification using adversarial training. For instance, Zhan *et al.* [54] have introduced a semisupervised HSI classification framework using a 1-D GAN for HSI (ssGAN1D). The ssGAN1D uses the discriminator features for the classification of HSI data. Zhu *et al.* [55] have used the GAN to generate new samples that are used for training the network for HSI classification. The spectral features are generated using a custom 1-D GAN in [56], which is used by another CNN for HSI classification with the help of majority voting. More recently, Feng *et al.* [57] have developed a multiclass spatial–spectral GAN (MSGAN) by employing two generators for spatial and spectral information, along with the adversarial objectives for multiple classes. Zhong *et al.* [58] introduce a semisupervised framework that combines the GAN with conditional random fields (CRFs) to address the data imbalance problem for HSI classification. A Caps-TripleGAN model has been investigated by Wang *et al.* [59] by generating the samples using a 1-D structure triple generative adversarial network (TripleGAN) and classifying the HSI data using the capsule network (CapsNet) [60]. Xue [61] has proposed a GAN-based HSI classification approach by utilizing a deep 3D-CNN-based generator network and another 3-D deep residual network-based discriminator network. The GAN has also been explored with CRFs for HSI classification [58]. The dense CRFs refine the generated HSI classification map. A DropBlock structure-based generative adversarial network (DBGAN) is investigated by Yin *et al.* [62] to generalize the HSI classification by randomly dropping the blocks. An adversarial training driven hallucination architecture is proposed by Pande *et al.* [63] to learn the features corresponding to the missing modalities for remote sensing. Wang *et al.* [64] have performed adversarial training to generate spectral samples and then used for HSI classification. The CapsNet and convolutional long short-term memory (ConvLSTM)-based discriminator network have been used by Wang *et al.* [65] to learn high-level contextual features for HSI classification.

Although the aforementioned GAN-based methods use adversarial training for HSI classification, they do not consider the minority classes. Generating the samples for each class or spectral band does not solve the class imbalance problem in HSI data sets. Recently, Mullick *et al.* [66] have introduced generative adversarial minority oversampling (GAMO) to generate the samples using a 1-D generator and a 1-D discriminator. Motivated by the problem of class imbalance in HSI data sets and the success of the GAMO model, in this article, we introduce a new `3D-HyperGAMO` model to deal with the class imbalance problem in HSI classification. The main contributions of this article can be summarized as follows.

1) We propose a new `3D-HyperGAMO` model to tackle the long-standing and most challenging problem of class imbalance for HSI data classification.
2) The proposed model generates a sufficient number of samples for minority classes by considering an oversampling strategy with the help of the convex 3-D patch generator, which is guided by another discriminator network following an adversarial learning manner.
3) The proposed method uses a conditional feature mapping unit and a convex 3-D patch generator (G) to generate the HSI patches within the boundary of the probability distribution of minority classes and a 3-D conditional discriminator (D) to distinguish between real and fake HSI samples.
4) To improve the feature representation ability, the proposed network architecture also uses a 3-D classifier network (M) to classify the 3-D HSI patches into the corresponding classes and achieves the state-of-the-art classification accuracy on four benchmark data sets using a very low amount of training samples.

The remainder of this article is organized as follows. Section II introduces the proposed `3D-HyperGAMO` model. Our experimental results are presented in Section III, along with a detailed discussion. Section IV concludes this article with some remarks and hints at plausible future research lines.

II. Proposed 3D-HyperGAMO Architecture

The HSI classification task suffers from the presence of imbalanced data for different classes. Generally, a percentage of the total data for each class is taken for the training of the model, which can be significantly affected by the class imbalance problem. In fact, most available HSI data sets are highly imbalanced, and training sets obtained by randomly selected samples are also imbalanced. This leads to a bias in the training of any CNN model toward the majority classes. In other words, the minority classes are not properly learned by the network due to the weak representation caused by the insufficient number of samples.

Here, we propose a `3D-HyperGAMO` model (inspired by the GAMO model [66]). The `3D-HyperGAMO` is introduced to oversample 3-D HSI patches from the minority classes, using a generative adversarial approach. Basically, our `3D-HyperGAMO` generates the new samples in the distribution of existing class-specific samples for minority classes from noise. The proposed `3D-HyperGAMO` model and all the attributes of the framework are illustrated in Fig. 2. The proposed framework consists of the following main components:

1) a 3-D-patch extractor (which takes full advantage of all the spectral information contained in the HSI data cube channels);
2) a conditional feature mapping unit to generate the intermediate features from noise;
3) a convex 3-D hyperspectral patch generator unit (PGU) to generate the new samples from the minority classes;
4) a 3-D conditional discriminator unit for the recognition of real versus fake 3-D patches;
5) a 3-D classifier network unit to perform the classification of the original and the generated 3-D HSI patches into categories.

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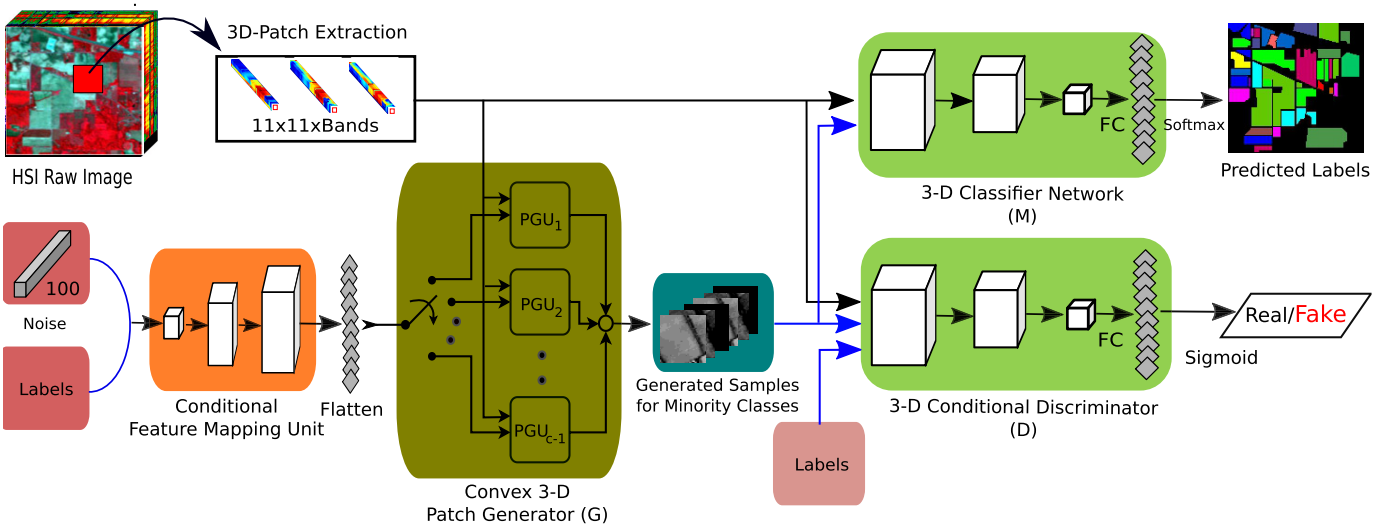


Fig. 2. Graphical overview of the proposed 3D-HyperGAMO architecture for GAMO when dealing with imbalanced HSI classification problems. It comprises a 3-D HSI patch extractor, a conditional feature mapping unit, a 3-D HSI patch generator to generate new data patches, a 3-D HSI conditional discriminator, and a 3-D HSI classification network.

A. 3-D-Patch Extractor

An HSI can be represented as a 3-D tensor with W, H, and B being the width, the height, and the number of spectral bands and defined as $\mathbf{X_{orig}} \in \mathcal{R}^{W\times H\times B}$. The HSI classification problem is generally performed by classifying all the pixel points into L_c land-cover classes, denoted by $Y = (y_1, y_2, \ldots, y_{L_c})$. The pixel $\mathbf{x}_{i,j} \in \mathbf{X_{orig}}$, where $i = 1, \ldots, W$ and $j = 1, \ldots, H$, is defined as a spectral vector $\mathbf{x}_{i,j} = [x_{i,j,1}, \ldots x_{i,j,B}] \in \mathcal{R}^B$, which represents the spectral signature of the captured surface materials. The 3-D HSI patches ($\mathbf{h}_i \in \mathcal{R}^{S\times S\times B}$) of dimension $S \times S \times B$ are extracted, where $S \times S$ is the spatial dimension and B is the number of channels. Basically, a sliding window approach with stride s is used over the spatial domain across each channel to generate the 3-D HSI patches that are treated as the input to a convex 3-D PGU and a 3-D classifier network. In this sense, each extracted HSI patch takes full advantage of all the spectral information contained in the HSI channels. To fulfill the desired goal of minority data oversampling using a deep learning-based framework (able to utilize both the spectral–spatial information effectively), we stack all the patches into $\mathbf{X} = \{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_i, \ldots, \mathbf{h}_k\}$, where $\mathbf{h}_i \in \mathcal{R}^{S\times S\times B}$ and k is the number of patches. In our experiments, a spatial window of size $S \times S \times B$ is defined (set empirically to $11 \times 11 \times$ Bands in the four considered data sets).

B. Conditional Feature Mapping Unit

This unit is used to generate the intermediate features (I_f) for the minority classes from a random vector (z). Consider $r^z \in \mathbb{R}_{\geq 0}$ as a random vector having z nonnegative real numbers and $l \in \{0, 1\}$ as a hot encoding binary vector with c values to represent the class labels, where c is the number of land-cover categories in the data set. Only one value of l is one (i.e., the one corresponding to the class for which we want to generate the new samples), and other values are zero. The aim of this unit is to map the random vector (r) to the intermediate features conditioned on the label vector (l). In this unit, first, the random vector (r) and, then, the label vector (l) are concatenated to form a combined vector having $z + c$ values. Then, a dense layer is used to generate the features, with length $256 * 7 * 7 = 12544$, followed by a LeakyReLU activation function and batch normalization using a momentum of 0.9. It is reshaped into a $256 \times 7 \times 7$ 3-D volume and then successively followed by a transpose of Conv2D layer to generate features of dimension $128 \times 5 \times 5$, with LeakyReLU activation function and batch normalization. The setting is used to generate the output of shape $1 \times 5 \times 5$ from the final transpose of the Conv2D layer but with linear activation. Thus, the final output of this unit is flattened to form an intermediate conditional feature vector (I_f) of length 784. Note that the output of the conditional feature mapping unit is connected to one of the convex 3-D PGUs based on the label (l) information. The detailed structure and the parameters of the conditional flatten feature mapping unit are shown in Table I.

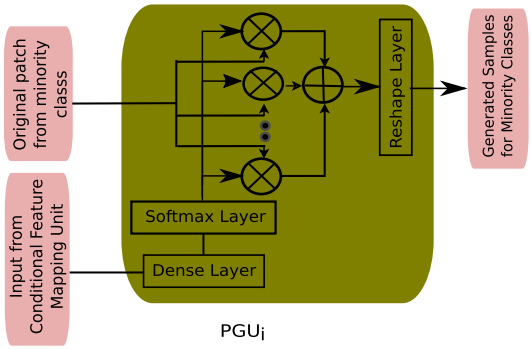


Fig. 3. Internal design architecture of a convex 3-D patch generation unit.

C. Convex 3-D PGU

The convex 3-D PGU is introduced to oversample the data points from the minority classes, as illustrated in Fig. 3. The class having the highest number of samples in the training set is considered as the majority class, whereas the remaining

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TABLE I
DETAILED STRUCTURE AND PARAMETERS OF 3D-HyperGAMO ARCHITECTURE

Conditional feature mapping unit				
Layer ID	**Kernel/Neurons**	**Stride**	**BatchNorm**	**Act. function**
Dense_1	256*7*7	-	Yes	LeakyReLU
Reshape_1	$256 \times 7 \times 7$	-	-	-
Conv2DTranspose_1	$128 \times 5 \times 5$	1	Yes	LeakyReLU
Conv2DTranspose_2	$64 \times 5 \times 5$	2	Yes	LeakyReLU
Conv2DTranspose_3	$1 \times 5 \times 5$	2	Yes	Linear
3D conditional discriminator				
Layer ID	**Kernel/Neurons**	**Stride**	**BatchNorm**	**Act. function**
Conv3D_1	$8 \times 1 \times 1 \times 7$	1	No	ReLU
Conv3D_2	$16 \times 3 \times 3 \times 5$	1	No	ReLU
Conv3D_3	$32 \times 5 \times 5 \times 7$	1	No	ReLU
ReshapeTo2D_1	-	-	-	-
Conv2D_1	$64 \times 3 \times 3$	1	No	ReLU
AveragePool_1	-	-	-	-
Dropout_1	0.5	-	-	-
Dense_1	1	-	-	Sigmoid
3D classifier network				
Layer ID	**Kernel/Neurons**	**Stride**	**BatchNorm**	**Act. function**
Conv3D_1	$8 \times 1 \times 1 \times 7$	1	No	ReLU
Conv3D_2	$16 \times 3 \times 3 \times 5$	1	No	ReLU
Conv3D_3	$32 \times 5 \times 5 \times 7$	1	No	ReLU
ReshapeTo2D_1	-	-	-	-
Conv2D_1	$64 \times 3 \times 3$	1	No	ReLU
AveragePool_1	-	-	-	-
Dropout_1	0.5	-	-	-
Dense_1	$n_classes$	-	-	Softmax

classes are considered as the minority classes. Thus, we use $c-1$ PGU units, one for each minority class. Note that c is the total number of classes in the given data set. $\text{PGU}_i|_{i\in[1,c-1]}$ refers to the PGU unit for the ith minority class. The PGU_i unit generates λ_i^g samples, such that

$$\lambda_i^g = \lambda_m - \lambda_i \tag{1}$$

for $\forall i \in [1, c-1]$, where λ_m is the number of training samples in the majority class and λ_i is the number of training samples in the ith minority class.

Any PGU unit takes two inputs, including the output of the conditional feature mapping unit (I_f having dimension: $784D$) and the 3D-patches of HSI data (dimension: $\lambda_i \times S \times S \times B$), where λ_i is the number of training patches from the ith class. At a given time, only one PGU unit is active. The PGU_i will be used when the 3D-Patch belongs to the ith minority class and when the ith binary bit (only) of the label's hot encoding vector (l) is set to one. A dense layer is used to transform the intermediate feature (I_f) into a feature of length λ_i, followed by the softmax activation function. This feature vector is repeated $\eta = S \times S \times B$ times to generate the class-specific random feature matrix (I_m) of dimension $\eta \times \lambda$. The input 3D-patches of the HSI data (dimension: $\lambda_i \times S \times S \times B$) are also reshaped into a matrix (P_m) of dimension $\lambda \times \eta$. Finally, the class-specific feature matrix (F_m) of dimension $\eta \times \lambda$ is generated as follows:

$$F_m = I_m \cdot (F_m)^T \tag{2}$$

where $(\cdot)$ is the dot product and $(F_m)^T$ is the transpose of matrix F_m. A flattened vector (F_v) of dimension (η) is computed by taking the columnwise sum for each row of matrix F_m. The vector F_v is reshaped in order to generate the new 3D-patch ($G_{3\text{dp}}$) of dimension $S \times S \times B$ for the ith class. Note that we refer to the generator network (G) as the combination of the conditional feature mapping unit and the 3-D PGU, as depicted in Fig. 2.

D. 3-D Conditional Discriminator Unit

We use a 3-D conditional discriminator (D) unit to differentiate between the original 3D-HSI patch and the generated 3D-HSI patch, using the generator network (G). This network tries to classify the original patch into a "real class" and the generated patch into a "fake class." This network consists of the following layers.

1) An input layer with original/generated 3D-HSI patch.
2) Three 3D-convolution layers with eight filters having kernel sizes of (1, 1, 7), followed by 16 filters having kernel sizes of (3, 3, 5) and followed by 32 filters having kernel sizes of (5, 5, 7).
3) A 3-D activation that is reshaped into a 2-D activation followed by a 2-D convolution layer with 64 filters having kernel sizes of (3, 3).
4) An average pooling layer followed by a dropout layer having 0.5 dropout factor leading to a final dense layer with a sigmoid activation function.
5) The final dense layer produces the probability of being classified as a real patch.

More information on the 3-D conditional discriminator network is given in Table I.

E. 3-D Classifier Network

The 3-D classifier network (M) plays the main role in the 3D-HyperGAMO network, which needs to be trained for HSI patch classification so that it can easily categorize the newly generated samples as validated by the discriminator. During the training of the classification network (M), the original training samples of each class (as well as the generated valid samples from the PGU of the minority classes) are used. During the testing stage, the test samples are used to evaluate the performance of the classifier network. Thus, the generator (G) and discriminator (D) networks are not used for testing. However, the use of generator (G) and discriminator (D) networks during training increases the learning capacity of the classifier (M) network by including more samples for the minority classes, generated within the boundary of those classes. The 3-D network architecture of the classifier network is the same as that of the 3-D discriminator network (up to the dropout layer). The difference is that, after that layer, the classifier network uses a dense layer having c outputs, followed by a softmax classifier layer. Basically, the final dense layer produces the class scores for each class, which are used to compute the softmax (cross-entropy) loss. A detailed summary is provided in Table I.

F. Objective Function

As mentioned earlier, our proposed method has three major networks: 3D-patch generator (G), 3D-patch discriminator (D), and 3D-patch classifier (M). The training is performed as a three-player minimax game, which is formulated as

$$\min_G \max_M \max_D \; O(G, M, D) = \sum_{i \in c} O_i \tag{3}$$

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where $O_i = (O_{i1} + O_{i2} + O_{i3} + O_{i4} + O_{i5} + O_{i6})$

$$O_{i1} = P_i \mathbb{E}_{\mathbf{h}\sim p_i^d}[\log M_i(\mathbf{h})]$$
$$O_{i2} = \sum_{j\in c\setminus\{i\}} P_j \mathbb{E}_{\mathbf{h}\sim p_j^d}[\log(1 - M_i(\mathbf{h}))]$$
$$O_{i3} = (P_c - P_i)\mathbb{E}_{G(\mathbf{z}|i)\sim p_i^g}[\log M_i(G(\mathbf{z}|i))]$$
$$O_{i4} = \sum_{j\in c\setminus\{i\}} (P_c - P_j)\mathbb{E}_{G(\mathbf{z}|j)\sim p_j^g}[\log(1 - M_i(G(\mathbf{z}|j)))]$$
$$O_{i5} = P_i \mathbb{E}_{\mathbf{h}\sim p_i^d}[\log D(\mathbf{h}|i)]$$
$$O_{i6} = (P_c - P_i)\mathbb{E}_{G(\mathbf{z}|i)\sim p_i^g}[\log(1 - D(G(\mathbf{z}|i)|i))]$$

where c is the number of classes, P_i is the prior probability of the ith class, p_i^g and p_i^d, respectively, denote the generated and real class conditional probability distributions of the ith class, and $\mathbf{h}$ and $G(\mathbf{z}|i)$, respectively, represent the original and generated 3D-HSI patches. Note that the aim of the generator (G) network is to fool the discriminator (O_{i6}). The purpose of the discriminator (D) network is to classify the original patch as real (O_{i5}) and the generated patch as fake (O_{i6}). The classifier (M) network maximizes the class scores for the ith class with respect to the original patch, as well as the generated patch of the ith class (O_{i1} and O_{i3}), and, at the same time, minimizes the class scores with respect to the other classes (O_{i2} and O_{i4}).

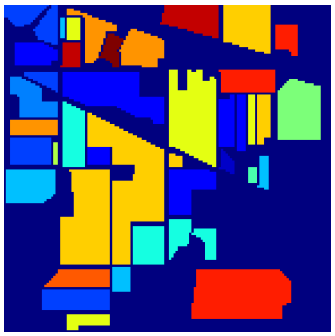
III. Experimental Results

Class imbalance is a long-standing problem in HSI classification, where classes do not provide equal numbers of representative samples to train the model. To show the effectiveness of the proposed method in solving this problem, we conduct classification experiments on four benchmark HSI data sets,[1] including Indian Pines (IP), Kennedy Space Center (KSC), University of Pavia (UP), and Botswana (BW). Disjoint train-test samples are used for IP and UP data sets. Here, disjoint means that, for these two data sets, the available labeled samples are already split into two subsets: those used for training and those used for testing. For the KSC and BW data sets, only labeled samples are available, without further separation into training or testing. This means that random samples need to be extracted for training and the remaining ones used for training. Figs. 4–7 show a detailed summary of these HSI data sets, including the ground truth, the type associated with the land-cover classes, and the number of available labeled samples per class.

A. Experimental Data Sets

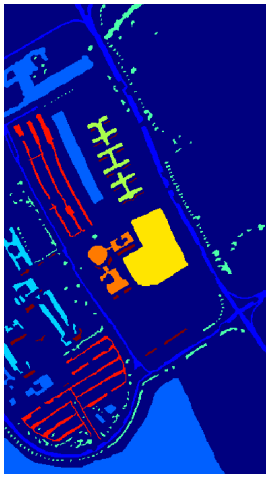
In the following, we describe the data sets used for experimental evaluation in our study.

1) The IP data set was gathered by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [67] over the IP test site in Northwestern Indiana. This data set contains images of 224 spectral bands within a wavelength range of 400–2500 nm. 24 null and corrupted bands have been removed. It consists of 16 mutually exclusive vegetation classes. The spatial dimensionality of the used scene is 145 × 145 pixels, and some of its classes contain highly imbalanced samples, as shown in Fig. 4.



Color	Land cover type	Samples
	Background	10776
	Alfalfa	46
	Corn notill	1428
	Corn min	830
	Corn	237
	Grass/Pasture	483
	Grass/Trees	730
	Grass/pasture-mowed	28
	Hay windrowed	478
	Oats	20
	Soybeans notill	972
	Soybeans min	2455
	Soybean clean	593
	Wheat	205
	Woods	1265
	Bldg Grass Tree Drives	386
	Stone steel towers	93
	Total samples	21025

Fig. 4. Ground truth, type associated with the land-cover classes, and the number of available samples in the IP data set.



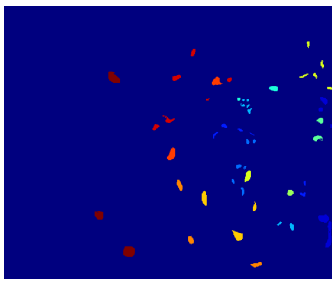
Color	Land cover type	Samples
	Background	164624
	Asphalt	6631
	Meadows	18649
	Gravel	2099
	Trees	3064
	Painted metal sheets	1345
	Bare Soil	5029
	Bitumen	1330
	Self Blocking Bricks	3682
	Shadows	947
	Total samples	207400

Fig. 5. Ground truth, type associated with the land-cover classes, and the number of available samples in the UP data set.

2) The KSC data set was gathered in 1996 by AVIRIS [67], with wavelengths ranging from 400 to 2500 nm. The image has 512 × 614 pixels and 176 spectral bands after removal of some low signal-to-noise ratio (SNR) bands. The KSC data set comprises 5202 labeled samples, with a total of 13 upland and wetland classes.
3) The UP data set was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor [68] during a flight campaign over the university campus at Pavia, Northern Italy. The data set contains nine land-cover classes from urban areas, and the spatial dimensionality is 610 × 340 pixels with 103 spectral bands in the wavelength range of 430–860 nm.
4) The BW data set was collected by the UT center for space research during 2001 to 2004 and acquired by Hyperion sensor on EO-1 over the Okavango, Delta, BW. The data set contains 14 land-cover classes from

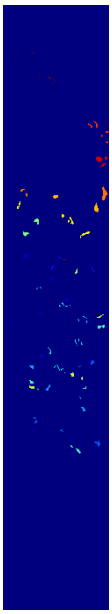
[1] http://dase.grss-ieee.org/

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Color	Land cover type	Samples
	Background	309157
	Scrub	761
	Willow swamp	243
	CP hammock	256
	Slash pine	252
	Oak/Broadleaf	161
	Hardwood	229
	Swap	105
	Graminoid marsh	431
	Spartina marsh	520
	Cattail marsh	404
	Salt marsh	419
	Mud flats	503
	Water	927
	Total samples	314368

Fig. 6. Ground truth, type associated with the land-cover classes, and the number of available samples in the KSC data set.



Color	Land cover type	Samples
	Background	374608
	Water	270
	Hippo grass	101
	Floodplain grasses1	251
	Floodplain grasses2	215
	Reeds1	269
	Riparian	269
	Firescar2	259
	island interior	203
	Accacia woodlands	314
	Accacia grasslands	248
	Short mopane	305
	Mixed mopane	181
	Exposed soils	268
	Total samples	377856

Fig. 7. Ground truth, type associated with the land-cover classes, and the number of available samples in the BW data set.

seasonal swamps, occasional swamps, and drier woodlands located in the distal portion of the Delta. The image comprises 1496×256 pixels having 30-m spatial resolution, with 242 spectral bands in the wavelength range from 400 to 2500 nm; 97 uncalibrated and water-corrupted bands have been removed, leading to 145 spectral bands.

B. Experimental Settings

In order to validate the effectiveness of the proposed `3D-HyperGAMO`, employed to oversample minority classes while training for HSI classification, we conduct a comparison that includes classical machine learning and representative deep learning methods available on [69],[2] such as random forest (RF) [70], multinomial logistic regression (MLR) [71], support vector machine (SVM) with radial basis function [72], gated recurrent unit (GRU) [73], long short-term memory (LSTM) [74], CNN-1D [75], CNN-2D [69], CNN-3D [23], [76], and the semisupervised ssGAN3D [54] model. The performance of the proposed `3D-HyperGAMO` is evaluated in terms of several widely used quantitative measurements, such as OA, AA, and statistical kappa (κ) coefficient [77]. The ratio of correctly classified samples among the total test samples is determined as OA. The mean of classwise accuracy is determined as AA. κ represents a strong mutual agreement between the generated classification maps of one network model and the provided ground truth. The weights of our `3D-HyperGAMO` network are randomly initialized, and the parameters of the model are optimized with the Adam optimizer [78], using a learning rate of 0.0002 for the IP, KSC, UP, and BW data sets. The spatial dimension of the noise vector is set to 100. To train the `3D-HyperGAMO` model, each experiment was run for 8000 iterations until the generalization ability of the generated training samples in the minority classes is stabilized. In our experiments, a spatial window of size $11 \times 11 \times B$ is extracted from the preprocessed IP, KSC, UP, and BW data sets, where B is related to the number of bands used for each data set. Table II shows the number of training samples (TRS) and synthesized samples (SYNS) generated during the training of the proposed `3D-HyperGAMO` network and the test samples (TES) used for the IP, UP, KSC, and BW data sets, respectively. The computer equipment used for experiments is composed of an Intel i9-9940X processor with 128 GB of DDR4 RAM, NVidia Titan RTX with 24 GB of DDR4 RAM. All the source code of the framework is implemented by using the Keras library with TensorFlow as the back end.

[2]https://github.com/mhaut/hyperspectral_deeplearning_review

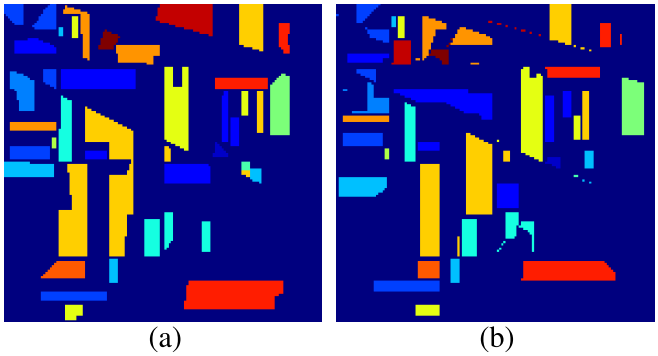


Fig. 8. Spatially disjoint training and test samples for the IP data set. (a) Disjoint train. (b) Disjoint test.

C. Classification Results

The widely used disjoint train-test for IP data set provided by the IEEE GRSS DASE (http://dase.grss-ieee.org/) is shown in Fig. 8, which contains highly imbalanced training samples. Thus, we use this setup to show the generalization ability of several HSI classification methods. Fig. 4 shows 16 land-cover classes and the number of associated samples per class for the disjoint IP data set. Table II shows the number of samples present in the 11th class of the IP data set, considered as the majority class. Since the class-specific samples are highly

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TABLE II

NUMBER OF TRAINING SAMPLES (TRS) AND SYNTHESIZED SAMPLES (SYNS) FOR TRAINING AND TEST SAMPLES (TES), OBTAINED FROM THE DISJOINT TRAIN-TEST IP DATA SET AND THE DISJOINT TRAIN-TEST UP DATA SET AND BY RANDOMLY SELECTING 5% TRAINING SAMPLES FROM THE KSC AND BW DATA SETS

Class	Disjoint Indian Pines			Disjoint University of Pavia			Kennedy Space Center			Botswana		
	TRS	SYNS	TES	TRS	SYNS	TES	TRS	SYNS	TES	TRS	SYNS	TES
1	29	1395	25	548	0	6304	38	462	723	14	486	256
2	762	662	675	540	8	18146	12	488	231	5	495	96
3	435	989	404	392	156	1815	13	487	243	13	487	238
4	146	1278	99	524	24	2912	13	487	239	11	489	204
5	232	1192	274	265	283	1113	8	492	153	13	487	256
6	394	1030	354	532	16	4572	11	489	218	13	487	256
7	16	1408	2	375	173	981	5	495	100	13	487	246
8	235	1189	250	514	34	3364	22	478	409	10	490	193
9	10	1414	10	231	317	795	26	474	494	16	484	298
10	470	954	503				20	480	384	12	488	236
11	1424	0	1065				21	479	398	15	485	290
12	328	1096	282				25	475	478	9	491	172
13	132	1292	80				46	454	881	13	487	255
14	728	696	545							5	495	90
15	291	1133	99									
16	57	1367	44									

TABLE III

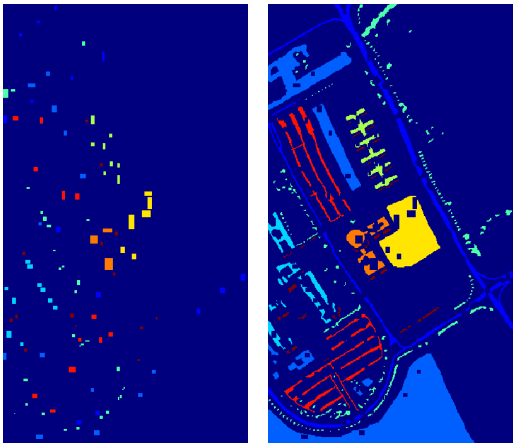
CLASSIFICATION RESULTS OBTAINED BY RF, MLR, SVM, GRU, LSTM, CNN1D, CNN2D, CNN3D, AND 3D-HyperGAMO ON THE DISJOINT TRAIN-TEST DATA SET FOR THE IP SCENE

Class	RF	MLR	SVM	MLP	GRU	LSTM	CNN1D	CNN2D	CNN3D	CNN3D+OV	ssGAN3D	Proposed
0	34.67	68.0	**96.0**	77.33	89.33	92.0	80.0	60.0	56.0	44.0	94.67	70.0
1	51.8	78.07	80.74	83.41	77.98	80.0	**83.75**	70.57	72.2	70.18	83.11	33.33
2	44.97	59.41	70.79	71.12	68.56	63.53	73.76	56.27	45.05	49.67	69.14	**97.33**
3	27.61	25.25	51.52	43.43	46.8	44.11	49.49	21.55	42.76	42.42	36.7	**74.24**
4	80.78	88.32	87.59	86.74	87.47	87.59	88.69	67.15	72.38	84.55	65.33	**96.67**
5	95.95	96.89	96.61	97.18	96.14	96.99	97.08	99.06	96.05	99.25	89.83	37.71
6	33.33	50.0	**100.0**	83.33	0.0	0.0	0.0	0.0	16.67	0.0	0.0	76.16
7	**99.87**	99.2	98.8	97.33	99.6	99.2	99.6	91.2	93.87	91.47	99.47	98.53
8	3.33	40.0	70.0	56.67	63.33	66.67	83.33	63.33	**90.0**	73.33	80.0	62.29
9	8.22	56.13	81.91	76.87	83.5	85.88	77.8	83.57	75.48	62.89	**88.54**	86.29
10	89.77	81.63	87.51	84.57	79.87	79.44	82.66	80.81	75.46	74.4	91.05	**96.52**
11	27.42	68.44	80.5	75.53	81.8	76.24	83.92	51.77	56.03	49.17	**91.49**	61.63
12	88.33	96.25	93.75	97.08	97.08	97.08	96.67	**100.0**	98.75	92.92	96.25	89.13
13	92.11	89.91	91.93	93.09	94.37	94.07	91.93	93.46	91.07	88.99	95.84	**98.23**
14	37.37	82.83	78.79	83.84	84.85	85.86	83.84	**90.24**	78.79	70.37	36.7	89.48
15	93.18	93.18	88.64	90.15	96.21	87.88	91.67	**98.48**	87.88	78.79	48.48	91.92
OA	65.9	78.15	85.08	83.91	83.36	82.96	84.62	77.65	75.17	73.39	84.57	**86.96**
AA	56.79	73.35	**84.69**	81.11	77.93	77.28	79.01	70.47	71.78	67.03	72.91	78.72
K(x100)	60.12	74.99	82.98	81.66	81.08	80.62	82.5	74.46	71.79	69.61	82.37	**85.17**

imbalanced, the lowest and highest numbers of synthesized samples are observed for the second and ninth classes of the disjoint IP data set (662 and 1414, respectively). Table III tabulates the classification performance of all the compared methods, including the OAs, AAs, and kappa coefficients, in addition to the class-specific accuracies for all the classes of the IP data set. The 16 rows in Table III represent the accuracy for each land-cover class, whereas the last three rows represent the OA, AA, and kappa values for all the compared methods. The highest classification results are represented in bold across all the compared methods. It can be observed that the proposed 3D-HyperGAMO model achieves better performance than RF, MLR, SVM, GRU, LSTM, CNN1D, CNN2D, CNN3D, CNN3D with standard oversampling (CNN3D + OV), and ssGAN3D methods, as shown in Table III, which shows the discriminative spectral and spatial feature learning capability of the 3-D convolutional layers of both the discriminator and the classifier. The proposed model achieves the highest OA (85.95%) and kappa coefficient (83.99%), while the SVM achieves the best AA (84.69%) performance for the IP data set. It is worth noting that the CNN1D and CNN2D both outperform the CNN3D compared with CNN, whereas the CNN1D achieves the best OA (84.62%), AA (79.01%), and kappa (82.50%) among these. To the best of our knowledge, this is probably due to the presence of very similar textures over most of the spectral bands. Hence, due to the increased redundancy among the spectral bands, the CNN1D and CNN2D seem to outperform the CNN3D, as observed in the experiments. Focusing on the CNN3D + OV, this model usually reaches the same accuracy results as CNN3D in some data sets, such as KSC and UP scenes, while, in the IP and BW, the results are slightly worse. This is quite interesting, as the standard oversampling usually copies the available samples randomly, i.e., it does not introduce new information. In this sense, standard oversampling can slightly improve the accuracy performance when the training data have been obtained uniformly, as in the KSC scene, or when the HSI samples are really easy to identify and separate into classes, as in the UP image (whose pixels are quite easy to identify by the classifier due to its relatively pure spectral signatures resulting from the high spatial resolution) and the rich spatial information present in the scene. However, this strategy will not produce better results when the data set is complex, such as the disjoint IP scene, where the spectral signatures are highly mixed and the spatial information is limited to borders between the different fields. In these cases, more elaborate sample selection strategies should be conducted. The proposed method is able to outperform these models in this context.

Among the traditional classifiers, RF, MLR, SVM, MLP, GRU, LSTM, and ssGAN3D, the best performance achieved is 85.08% (OA), 84.69% (AA), and 82.98% (Kappa) for ssGAN3D trained using newly generated synthesized samples. During the training phase, the newly generated samples can properly represent the characteristics of minority classes and provide helpful information to learn better the weights due to the combined efforts of the conditional feature mapping unit and the convex patch generator. The above observation may be one of the reasons behind the performance improvement observed after using the proposed oversampling-based HSI classification model. In addition, the proposed 3D-HyperGAMO model exhibits a significant improvement of at least 0.87% in OA and 1.01% in kappa values compared to other methods listed in Table III.

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(a) (b)

Fig. 9. Spatially disjoint training and test samples for the UP data set. (a) Disjoint train. (b) Disjoint test.

TABLE IV

CLASSIFICATION RESULTS OBTAINED BY RF, MLR, SVM, GRU, LSTM, CNN1D, CNN2D, CNN3D, AND 3D-HyperGAMO ON THE DISJOINT TRAIN-TEST DATA SET FOR THE UP SCENE

Class	RF	MLR	SVM	MLP	GRU	LSTM	CNN1D	CNN2D	CNN3D	CNN3D+OV	ssGAN3D	Proposed
0	79.62	77.69	82.23	84.06	79.27	79.3	87.55	81.71	84.91	83.48	76.61	**97.4**
1	55.3	58.78	65.81	73.42	89.61	77.42	88.23	93.75	92.36	93.58	88.44	**99.49**
2	45.29	67.22	66.72	67.6	51.55	58.35	75.81	64.59	65.6	64.3	72.97	**91.71**
3	**98.68**	74.29	97.77	92.26	90.13	98.04	92.63	96.25	97.68	97.22	96.25	71.31
4	99.13	98.89	99.37	99.46	99.49	99.1	99.67	**99.79**	98.2	99.4	98.68	98.3
5	78.61	93.53	91.62	92.06	34.39	65.05	87.58	67.56	76.6	75.62	85.1	**97.99**
6	79.31	85.12	87.36	89.26	78.12	88.07	88.01	74.96	81.69	83.83	74.38	97.25
7	90.9	87.59	90.46	91.26	83.59	91.15	89.01	**97.29**	95.47	93.83	93.79	94.1
8	97.57	99.21	93.71	99.16	95.68	99.33	**99.33**	92.33	96.6	96.85	90.86	92.09
OA	70.14	72.23	77.8	81.46	79.59	79.39	88.41	87.69	88.8	88.88	86.5	**93.9**
AA	80.49	82.48	86.12	87.62	77.98	83.98	89.76	85.36	87.68	87.57	86.34	**93.29**
K(x100)	62.98	65.44	72.06	76.23	72.14	73.14	84.63	83.35	84.97	85.02	82.14	**91.86**

To determine the generalization power of the considered HSI classification methods, the highly imbalanced UP disjoint train-test set is also adopted and shown in Fig. 9. Fig. 5 shows nine land-cover classes and the number of samples belonging to each class for this data set. Table II shows that the number of synthesized samples generated during the training of the proposed network for the disjoint UP data set is less compared to the disjoint IP data set. The minimum number of samples (947) is present in the Shadows class, while the Meadows class contains the maximum number of samples (947) in the UP data set. The OA, AA, and kappa coefficients are shown in Table IV along with the class-specific accuracies for all the land cover classes of the UP data set. Table IV shows the superior performance of the proposed 3D-HyperGAMO model in terms of OA (92.46%), AA (90.87%), and kappa (89.87). When considering the UP data set, the performance of the CNN3D is better, and we observe that the second highest performance is obtained by using CNN3D with OA (88.88%) and kappa (84.90%) in Table IV. The proposed 3D-HyperGAMO model achieves significant performance improvements of at least 3.58%, 4.47%, and 4.97% in terms of OA, AA, and kappa compared with the CNN3D reported in Table IV. The SVM, a traditional classifier, achieves good performance in terms of OA (77.80%), AA (86.12%), and kappa (72.06%) compared to the RF (70.14%, 80.49%, and 62.98%). It can also be observed that the MLP performs quite well compared with GRU and LSTM. It is worth mentioning that ssGAN3D provides significantly better OA (86.50%) compared to the classical learning models. Overall, the conducted oversampling on minority classes allows the proposed model to learn more relevant features distributed across different categories.

TABLE V

CLASSIFICATION RESULTS OBTAINED BY RF, MLR, SVM, GRU, LSTM, CNN1D, CNN2D, CNN3D, AND 3D-HyperGAMO USING 5% RANDOMLY SELECTED TRAINING SAMPLES FROM THE KSC DATA SET

Class	RF	MLR	SVM	MLP	GRU	LSTM	CNN1D	CNN2D	CNN3D	CNN3D+OV	ssGAN3D	Proposed
0	94.24	95.44	94.33	93.82	95.62	95.94	96.86	98.16	**98.8**	98.11	97.83	97.33
1	74.17	82.68	**92.21**	85.86	87.01	80.09	89.47	86.0	77.78	83.4	80.38	71.68
2	86.56	75.03	81.48	91.5	85.6	83.81	91.08	**96.02**	88.61	94.51	82.72	88.84
3	60.39	58.16	62.9	54.53	49.09	49.93	76.71	50.77	57.88	53.84	67.78	**89.9**
4	46.19	60.35	55.34	49.89	61.44	42.27	59.48	67.76	66.45	69.93	52.72	**82.99**
5	41.47	57.76	61.29	55.3	54.22	54.99	66.97	**94.47**	88.48	92.78	82.11	77.82
6	65.0	90.33	68.67	75.0	84.0	74.33	81.0	93.67	86.67	96.0	81.0	**99.74**
7	67.48	90.71	89.98	89.81	87.53	86.39	86.55	95.11	90.3	87.2	88.02	**99.58**
8	88.19	94.67	90.35	95.61	95.55	91.3	98.45	**99.12**	97.71	96.49	83.87	94.95
9	75.61	91.49	93.23	89.84	90.89	82.81	92.53	97.57	93.75	94.01	97.83	**99.37**
10	93.8	95.47	96.65	92.71	97.57	92.46	93.05	**99.58**	99.5	95.98	97.74	98.85
11	77.96	93.65	91.07	86.61	88.21	89.47	89.89	96.03	96.37	95.96	95.12	**98.3**
12	99.7	**100.0**	99.96	99.51	100.0	99.39	99.39	100.0	100.0	100.0	100.0	100.0
OA	82.01	89.07	88.94	87.78	88.55	85.93	90.92	94.04	92.48	92.54	90.31	**95.31**
AA	74.67	83.52	82.88	81.54	82.83	78.71	86.26	90.33	87.87	89.09	85.16	**92.26**
K(x100)	79.92	87.83	87.68	86.38	87.24	84.31	89.87	93.36	91.62	91.69	89.21	**94.78**

The performance of the proposed 3D-HyperGAMO model has been also evaluated using the KSC data set. The detailed summary of land-cover classes in the KSC data set is shown in Fig. 6. The number of class-specific samples and the difference between the minimum and maximum classwise samples show that the KSC data set contains mild imbalances compared to the IP and UP data sets. In order to show the effectiveness of the proposed 3D-HyperGAMO network, the labeled samples in the KSC data set are randomly divided into 5% for training and 95% for testing, respectively. Table II shows that the number of samples in the majority class of the training set is too small to achieve model generalization for the KSC data set. Thus, it is difficult to train the models, even with the samples after minority oversampling during the training of the 3D-HyperGAMO, leading to poor generalization power. To improve the generalization capability of the proposed model, the same number of samples are synthesized for all the classes of the KSC data set depending on the value of a hyperparameter $resSamplePd$, which is set to 500. The statistical analyses in terms of the OA, AA, and kappa metrics are reported in Table V, and the best results are highlighted in bold. The 3D-HyperGAMO model obtains the highest classification accuracies in terms of all considered measures, i.e., OA (95.19%), AA (90.94%), and kappa (94.64%). Among the different CNN models, CNN2D exhibits the highest performance improvement in terms of OA (1.56%), AA (2.46%), and kappa (1.74%) over the CNN3D. The performance of SVM is always commendable compared with classical machine learning-based methods, i.e., RF, but its performance is a bit lower than that achieved by the MLR. Due to the presence of fewer imbalances in the samples, MLP shows great improvement using OA, AA, and kappa over GRU and LSTM. It can also be observed in Table V that ssGAN3D shows better performance than classical methods but worse than all CNN models. Moreover, the proposed network shows significant performance gains of at least 1.15% in OA, 0.61% in AA, and 1.28% in kappa values compared to the methods listed in Table V.

The performance of the proposed 3D-HyperGAMO network is also evaluated in the high-resolution BW HSI data set. Fig. 7 shows the description of the land-cover classes and the number of available samples per class, which reveals imbalances for

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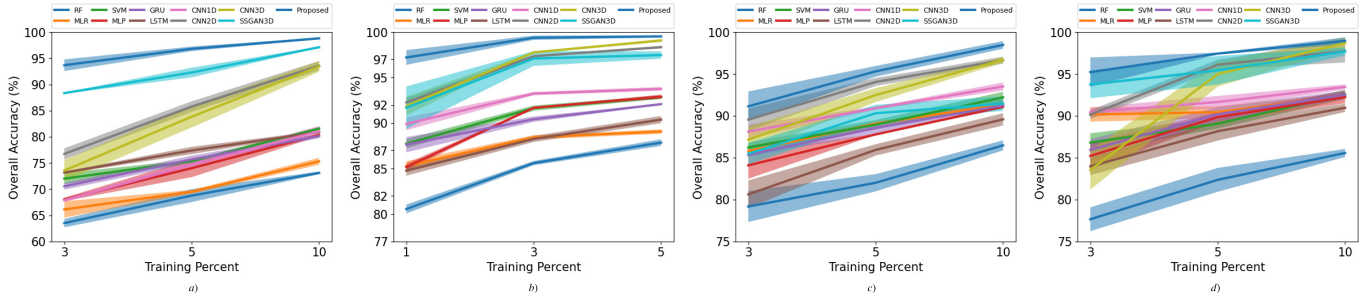


Fig. 10. OA obtained by different methods with different training percentages. (a) IP data set. (b) UP data set. (c) KSC data set. (d) BW data set.

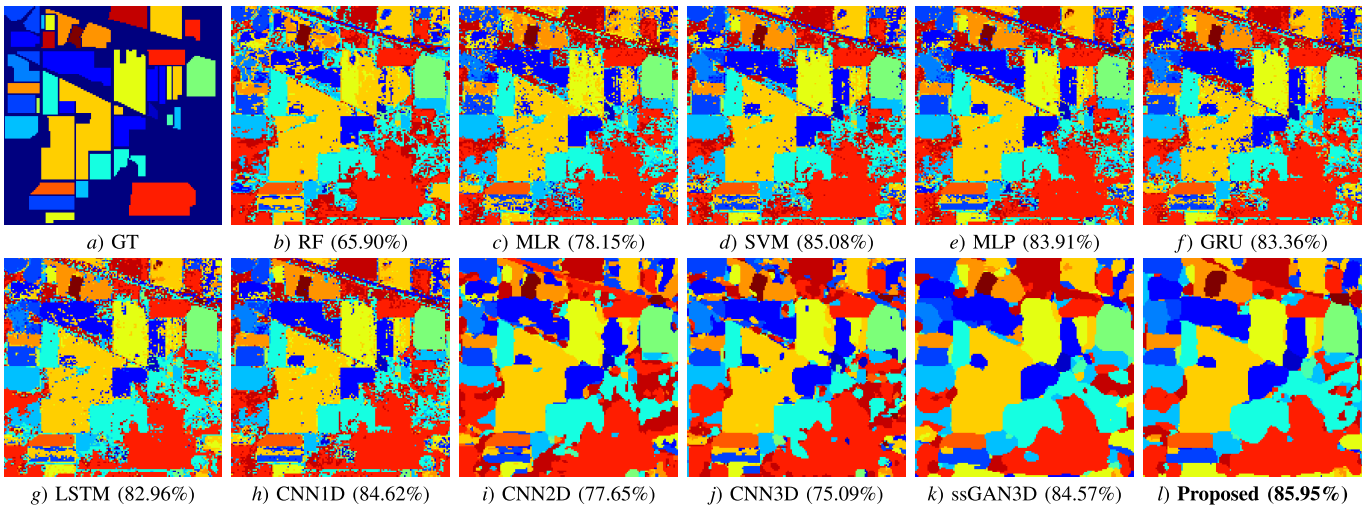


Fig. 11. (a) Ground truth and classification maps obtained for the disjoint IP data set by (b) RF, (c) MLR, (d) SVM, (e) MLP, (f) GRU, (g) LSTM, (h) CNN1D, (i) CNN2D, (j) CNN3D, (k) ssGAN3D, and (l) 3D-HyperGAMO models, using the fixed training set.

TABLE VI

CLASSIFICATION RESULTS OBTAINED BY RF, MLR, SVM, GRU, LSTM, CNN1D, CNN2D, CNN3D, AND 3D-HyperGAMO USING 5% RANDOMLY SELECTED TRAINING SAMPLES FROM THE BW DATA SET

Class	RF	MLR	SVM	MLP	GRU	LSTM	CNN1D	CNN2D	CNN3D	CNN3D+OV	ssGAN3D	Proposed
0	99.61	**100.0**	99.87	100.0	100.0	99.48	100.0	98.96	99.35	99.74	99.48	99.31
1	86.81	96.18	96.18	95.83	89.93	82.99	80.56	95.83	98.26	100.0	90.28	97.04
2	92.58	98.18	95.24	96.78	97.34	97.06	**99.3**	98.88	97.9	98.32	95.1	98.45
3	86.11	92.49	91.99	91.67	93.95	95.26	98.2	99.18	91.67	91.5	100.0	91.02
4	75.88	82.9	87.08	80.94	78.65	77.29	84.6	82.16	82.37	81.12	79.95	**98.04**
5	48.29	69.36	66.49	77.71	75.69	71.84	74.55	93.86	88.01	87.71	93.49	**100.0**
6	91.73	95.8	96.48	95.8	95.39	95.66	97.43	99.32	99.73	99.86	96.34	99.72
7	83.94	98.1	91.02	97.93	98.62	94.13	**99.65**	97.41	96.72	91.71	93.78	88.8
8	77.85	83.67	80.65	86.13	85.68	79.31	88.93	**98.1**	92.51	93.29	96.98	93.23
9	75.56	87.29	85.73	82.06	82.63	83.62	90.4	**99.86**	94.77	97.88	99.15	99.19
10	88.28	92.53	90.34	93.33	91.26	92.53	89.89	96.44	98.74	97.82	99.31	**100.0**
11	87.98	92.25	92.25	93.8	95.93	93.8	94.77	99.81	97.87	99.22	96.51	**100.0**
12	77.12	92.55	89.93	85.62	89.8	83.53	93.33	96.86	99.87	99.74	98.69	**99.89**
13	96.67	96.3	97.04	84.07	98.15	95.19	87.41	81.48	95.19	91.11	91.85	**98.99**
OA	82.36	90.43	89.07	89.84	90.19	88.16	91.65	96.1	94.99	94.83	95.43	**97.43**
AA	83.46	91.25	90.02	90.12	90.93	88.69	91.36	95.58	95.21	94.93	95.06	**97.4**
K(x100)	80.89	89.63	88.16	89.0	89.37	87.17	90.95	95.77	94.57	94.41	95.05	**97.22**

this data set (as it was already the case with KSC). To test the performance of the proposed 3D-HyperGAMO network, the BW data set is randomly divided into training and test sets, where 5% of the available samples are utilized for training and the remaining 95% are used for testing. Table II shows the number of synthesized samples (SYNS) for the BW data set, generated after randomly selecting 5% of the available labeled samples for the training of 3D-HyperGAMO. Table VI reports the achieved classwise accuracy and the statistical analysis of parameters: OA, AA, and kappa for all the compared methods. The proposed network achieves the highest classification performance on the basis of OA (98.45%), AA (98.56), and kappa (98.33) compared to the methods reported in Table VI. The CNN2D performs well on the BW data set compared to CNN1D and CNN3D while achieving an OA of 96.10%, an AA of 95.58%, and a kappa of 95.77% during the model evaluation. ssGAN3D achieves better OA (95.43%) performance compared to CNN1D (91.65%) and CNN3D (94.99%), respectively. The performance achieved using standard classifiers show that the MLR performs quite well compared with RF, SVM, and MLP. It is worth noting that, among the recurrent classifiers, GRU exhibits better performance improvements compared to LSTM. The proposed 3D-HyperGAMO model achieves a performance gain of at least 2.35% in OA, 3.01% in AA, and 2.56% in kappa compared to the methods listed in Table VI.

D. Performance Evaluation With Different Training Percentages

In order to illustrate the robustness and generalization ability of the proposed 3D-HyperGAMO network, it is important to justify the performance achieved on small and varying training sets. Our experiments are conducted overtraining sets with different percentages of randomly selected samples. Specifically, we randomly select 3%, 5%, and 10% samples from IP, KSC, and BW data sets, whereas 1%, 3%, and 5%

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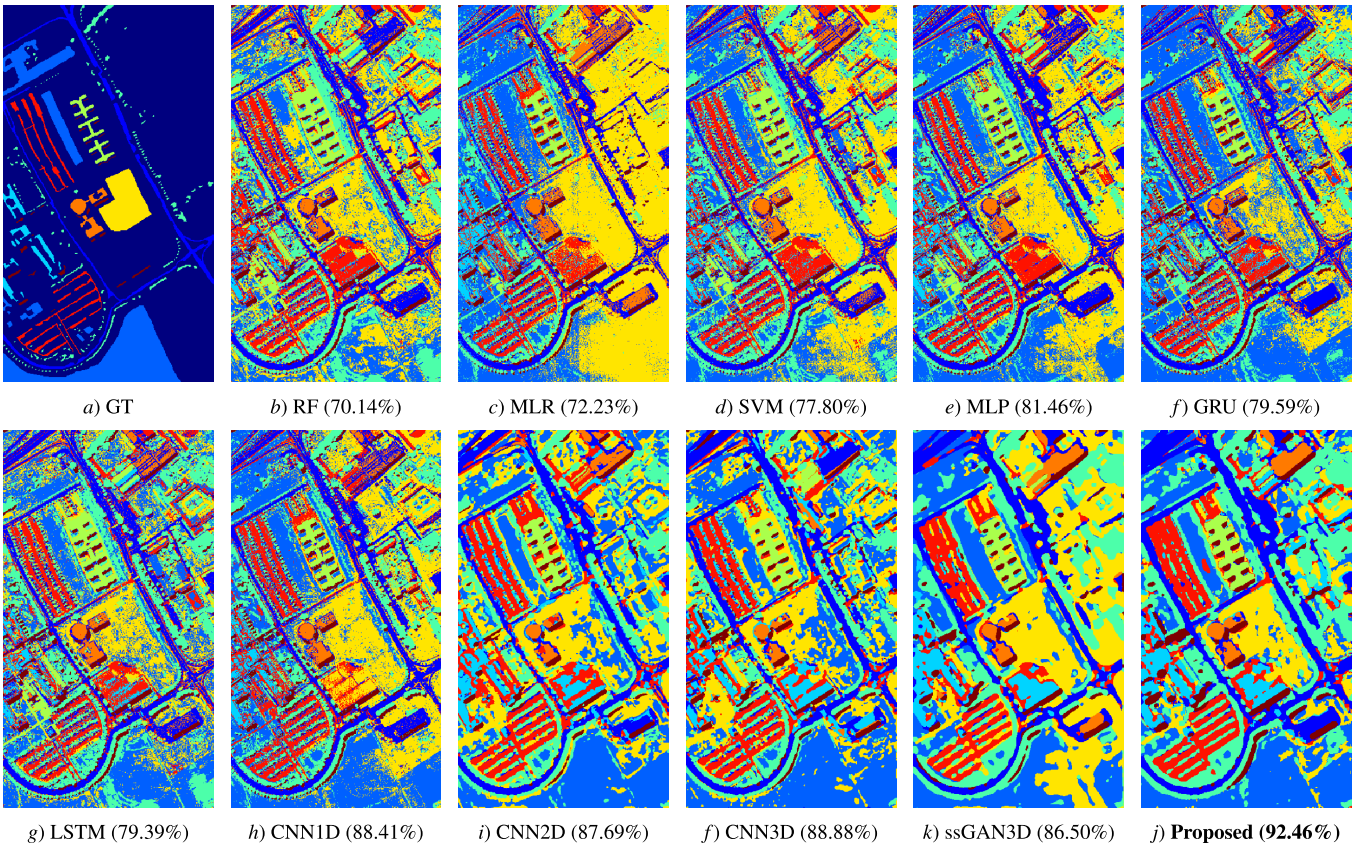


Fig. 12. Classification maps obtained for the disjoint UP data set by (a) RF, (b) MLR, (c) SVM, (d) MLR, (e) GRU, (f) LSTM, (g) CNN1D, (h) CNN2D, (i) CNN3D, and (j) `3D-HyperGAMO` models, using the fixed training set.

randomly selected samples have been selected from the UP data set. Fig. 10 illustrates the achieved OA against different training sizes using the different compared methods for the IP, UP, KSC, and BW data sets. It can be easily seen that the gain in performance using the proposed method is more significant for the data sets with a higher degree of class imbalance.

Fig. 10(a) shows that the proposed `3D-HyperGAMO` network consistently outperforms the other compared methods using varying training samples, with significant differences for the IP data set. The performance of CNN1D is not so good, but ssGAN3D achieves significantly better performance compared to CNN2D and CNN3D and similar performance for larger training sizes compared to CNN2D and CNN3D. In addition, SVM, MLP, and LSTM exhibit similar performance when the training size increases, but the improvement in performance is not comparable for the RF. Fig. 11(a)–(h) shows the results of the classification methods: RF, MLR, SVM, MLP, GRU, LSTM, CNN1D, CNN2D, CNN3D, and the proposed method for the IP data set. It can be observed that, even with very few samples, the proposed `3D-HyperGAMO` improves the obtained classification maps compared to other methods.

Fig. 10(b) demonstrates the significant improvement in OA performance achieved by the proposed network compared to the other tested methods when considering a small training set for the UP data set. The CNN2D outperforms the CNN3D when the training size is very small, but, once the training size becomes more than 3%, the CNN3D improves the performance compared with the CNN2D and ssGAN3D. Among all the methods, RF exhibits the worst performance, while the MLP and SVM achieve similar OA for larger training sizes. Similarly, LSTM performs better than GRU.

Fig. 10(c) illustrates the achieved performance in terms of OA (under varying training sizes) for the KSC data set. It is evident that the proposed `3D-HyperGAMO` network smoothly outperforms the other obtained methods. In the beginning, the CNN2D performs well compared to the CNN3D but then achieves similar OA when the size of the training set increases. The MLR, MLP, and ssGAN3D obtain very similar performance. The LSTM achieves comparable performance in relation to the RF.

Fig. 10(d) illustrates the performance gained in terms of the OA under varying training sizes for the BW data set. The proposed `3D-HyperGAMO` network achieves outstanding performance compared with other methods. Even though the ssGAN3D and CNN2D perform well for small training sizes, the CNN3D obtains the best performance for a higher number of training samples. It is also worth noting that the RF provides OA results that are not at the same quality level throughout the experiments. The MLR, MLP, and SVM achieve similar performance in terms of OA values, with increasing training samples, but the worst OA is achieved by the LSTM.

E. Discussion of the Obtained Classification Maps

The robustness of any HSI classification method can also be verified through an evaluation of the generated classification

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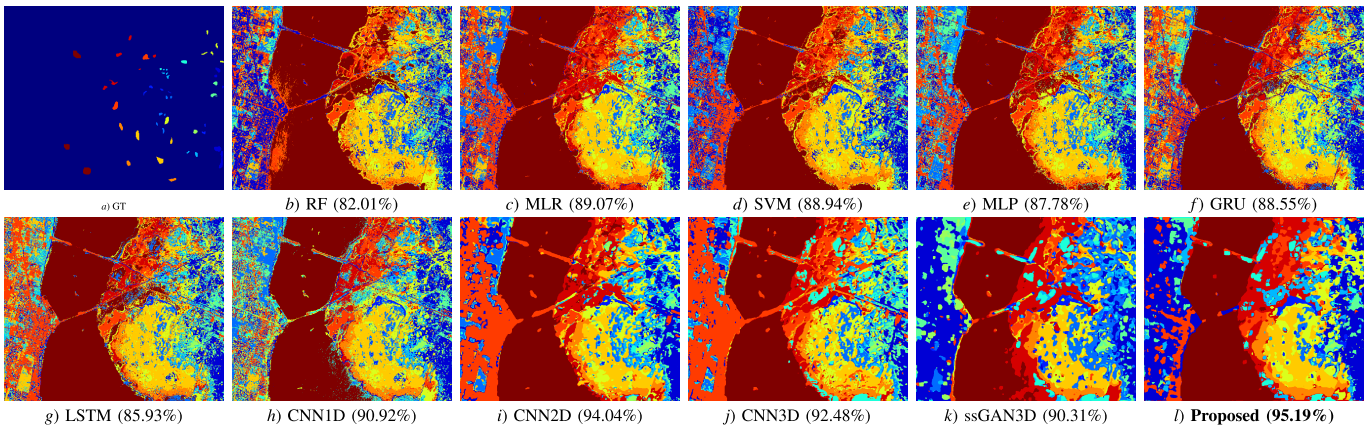


Fig. 13. (a) Ground truth and classification maps obtained for the KSC data set by (b) RF, (c) MLR, (d) SVM, (e) MLP, (f) GRU, (g) LSTM, (h) CNN1D, (i) CNN2D, (j) CNN3D, (k) ssGAN3D, and (l) 3D-HyperGAMO models, using 5% randomly selected training samples.

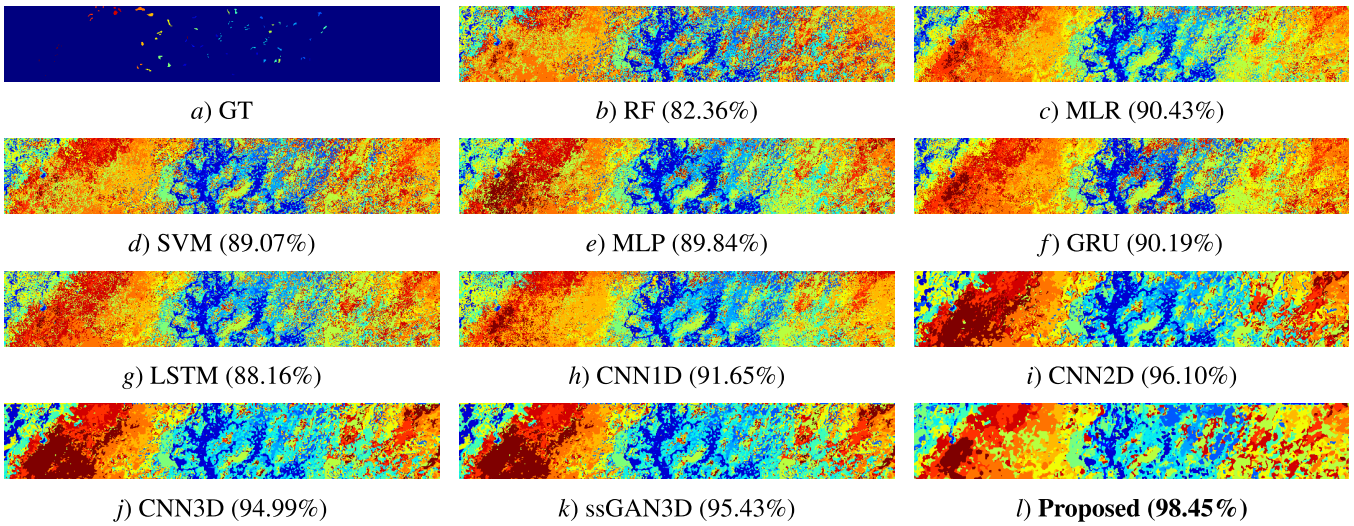


Fig. 14. (a) Ground truth and classification maps obtained for the BW data set by (b) RF, (c) MLR, (d) SVM, (e) MLP, (f) GRU, (g) LSTM, (h) CNN1D, (i) CNN2D, (j) CNN3D, (k) ssGAN3D, and (l) 3D-HyperGAMO models, using 5% randomly selected training samples.

maps. Figs. 11–14 show the classification maps obtained by the nine compared methods on the IP, UP, KSC, and BW data sets, respectively. As shown in Fig. 11(b)–(h), the maps for RF, MLR, SVM, MLP, GRU, LSTM, and CNN1D contain salt and pepper noise in the classification results due to the misclassification of many points in the center of land-cover regions. This is clearly visible in the land-cover classes, such as Corn min, Corn, Hay windrowed, and Building Grass Tree Drives, in the generated map of the IP data set. Compared to the previous methods, the CNN2D, CNN3D, and ssGAN3D obtain classification maps with higher quality, whereas the proposed 3D-HyperGAMO model significantly boosts the performance through the generated classification maps. Moreover, it also helps to improve the uniformity of the land-cover regions shown in Fig. 11(h)–(l). The generated maps for the UP data set using RF, MLR, SVM, MLP, GRU, LSTM, CNN1D, CNN2D, CNN3D, ssGAN3D, and 3D-HyperGAMO methods are depicted in Fig. 12. The presence of similar spectral classes leads to an increase in the misclassification probability. However, the proposed method can successfully distinguish the different classes. Moreover, 3D-HyperGAMO achieves better delineation in almost all class boundaries compared to other methods.

Fig. 13 gives a comparison of the classification maps using various methods for the KSC data set. Fig. 13(b)–(i) shows that the obtained maps contain salt and pepper noise. Compared to the CNN1D, CNN2D, and ssGAN3D, the CNN3D generates better classification maps. The proposed 3D-HyperGAMO network obtains significant qualitative improvements compared to other maps. A very similar trend is also observed for the BW data set, as illustrated in Fig. 14. In other words, the classification maps generated using the proposed method are qualitatively better than those obtained by other methods.

IV. Conclusion

In this article, we proposed a new 3D-HyperGAMO framework that serves as an effective minority class oversampling technique for handling the class imbalance in HSI data classification problems. It consists of a 3D-generator network that takes the 3D-HSI patches and a middle-level representation of a noise vector produced by a conditional feature mapping unit to generate the new samples of that class. It also consists

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of a 3-D conditional discriminator that distinguishes between the original and generated samples. In order to learn the class-specific information, a 3-D classification network is also used—which classifies the samples (original and generated) into the corresponding classes. The number of samples to be generated is decided by the difference between the number of samples in the majority class and those in the corresponding minority class. The proposed method has been tested on four benchmark HSI data sets. The results have been compared with traditional and CNN-based methods. The proposed model outperforms existing models in most of the cases. Moreover, the performance gain obtained by the proposed model is more significant for small training sets. It can also be noticed that the performance of the proposed model is significantly improved when the degree of imbalance is higher. The classification maps generated using the proposed model are smoother and perceptually more appealing compared to those obtained by the other tested models. Thus, from an experimental viewpoint, we can conclude that the proposed `3D-HyperGAMO` is more effective with small training sets with high imbalance, which is the most common scenario in real remote sensing applications.

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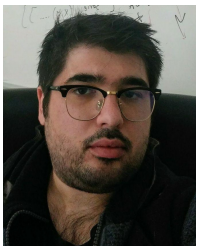
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